



Tree diagrams

TREE DIAGRAMS CAN BE CONSTRUCTED TO HELP CALCULATE THE PROBABILITY OF MULTIPLE EVENTS OCCURRING.

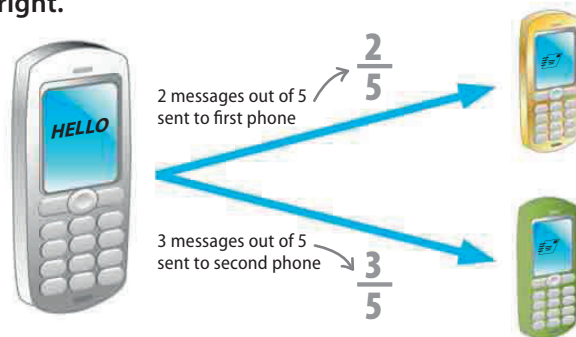
A range of probable outcomes of future events can be shown using arrows, or the “branches” of a “tree,” flowing from left to right.

Building a tree diagram

The first stage of building a tree diagram is to draw an arrow from the start position to each of the possible outcomes. Here, the start is a cell phone, and the outcomes are 5 messages sent to 2 other phones, with each of these other phones at the end of 1 of 2 arrows. Because no event came before, they are single events.

▷ Single events

Of 5 messages, 2 are sent to the first phone, shown by the fraction $\frac{2}{5}$, and 3 out of 5 are sent to the second phone, shown by the fraction $\frac{3}{5}$.



SEE ALSO

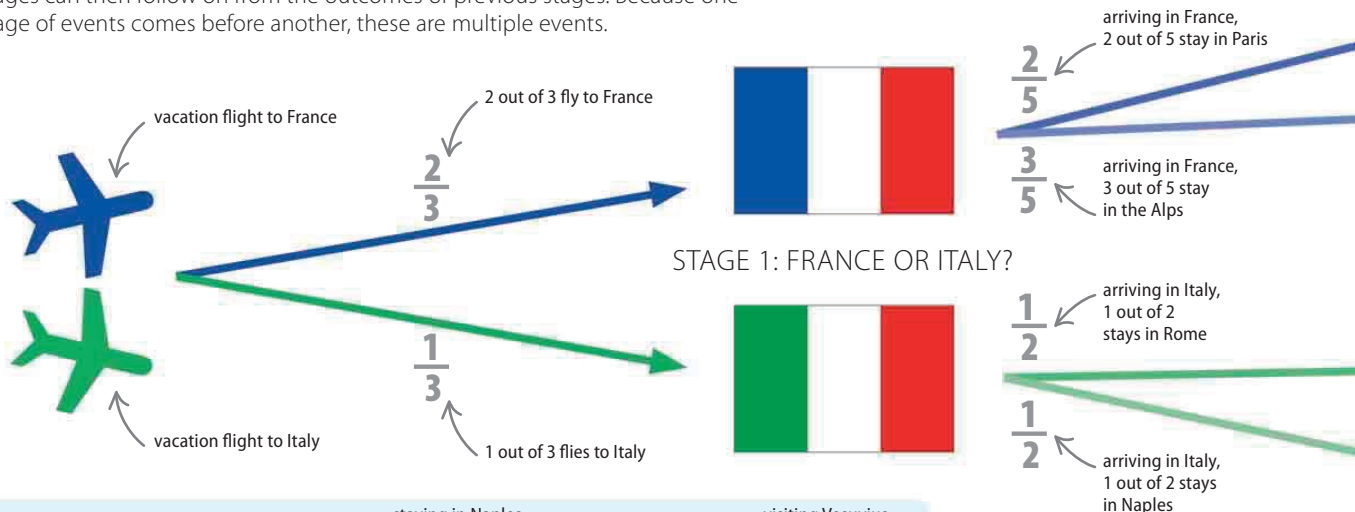
◀ 230–231 What is probability?

◀ 234–235 Combined probabilities

◀ 236–237 Dependent events

Tree diagrams showing multiple events

To draw a tree diagram that shows multiple events, begin with a start position, with arrows leading to the right to each of the possible outcomes. This is stage 1. Each of the outcomes of stage 1 then becomes a new start position, with further arrows each leading to a new stage of possible outcomes. This is stage 2. More stages can then follow on from the outcomes of previous stages. Because one stage of events comes before another, these are multiple events.



Find the probability

To work out the chance of a randomly selected person flying to Italy, staying in Naples, and visiting Vesuvius, multiply the chances of each stage of this trip together for the answer.

$$\frac{1}{3} \times \frac{1}{2} \times \frac{1}{4} = \frac{1}{24}$$

staying in Naples
 1 out of 3 flies to Italy
 chance of person visiting Italy, then Naples, then Vesuvius

△ Multiple events in 3 stages

The tree diagram above shows 3 stages of a vacation. In stage 1, people fly to France or Italy.